

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1269

Greenway 138 kV
October 16, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1269. The Interconnection Customer proposes a 261.9 MW solar photovoltaic facility interconnecting at the Greenway 138 kV substation in the Virginia area. The request was received on August 17, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 452 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,900,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: PASCO CNTY SOLID WASTE RESOURCE RECOVERY to NOVEC PARADISE 138 kV
Contingency: PASCO CNTY SOLID WASTE RESOURCE RECOVERY to VMEA EAKING GEN 138 kV line
Loading: 118.5% (Rating: 245 MVA)
DFAX: 6.5%
Pre-project loading: 111.6%
2. Facility: YARDLEY RIDGE to GREENWAY 138 kV
Contingency: JESSUP to POLAND RD SUBSTATION 138 kV
Loading: 114.4% (Rating: 209 MVA)
DFAX: 67.9%
Pre-project loading: 29.2%
3. Facility: PASCO CNTY SOLID WASTE RESOURCE RECOVERY to VEA PEAKING GEN 138 kV line
Contingency: PASCO CNTY SOLID WASTE RESOURCE RECOVERY to NOVEC PARADISE 138 kV line
Loading: 108.3% (Rating: 252 MVA)
DFAX: 7.0%
Pre-project loading: 101.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PASCO CNTY SOLID WASTE RESOURCE RECOVERY to NOVEC PARADISE 138 kV Line	Rebuild line with higher-capacity conductor	\$2,684,000	41% (\$1,089,000)
YARDLEY RIDGE to GREENWAY 138 kV	Rebuild line with higher-capacity conductor	\$7,440,000	74% (\$5,508,000)
VMEA PEAKING GEN to PASCO CNTY SOLID WASTE RESOURCE RECOVERY 138 kV Line	Reconductor line	\$953,000	32% (\$303,000)
	Total allocated to project		\$6,900,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.